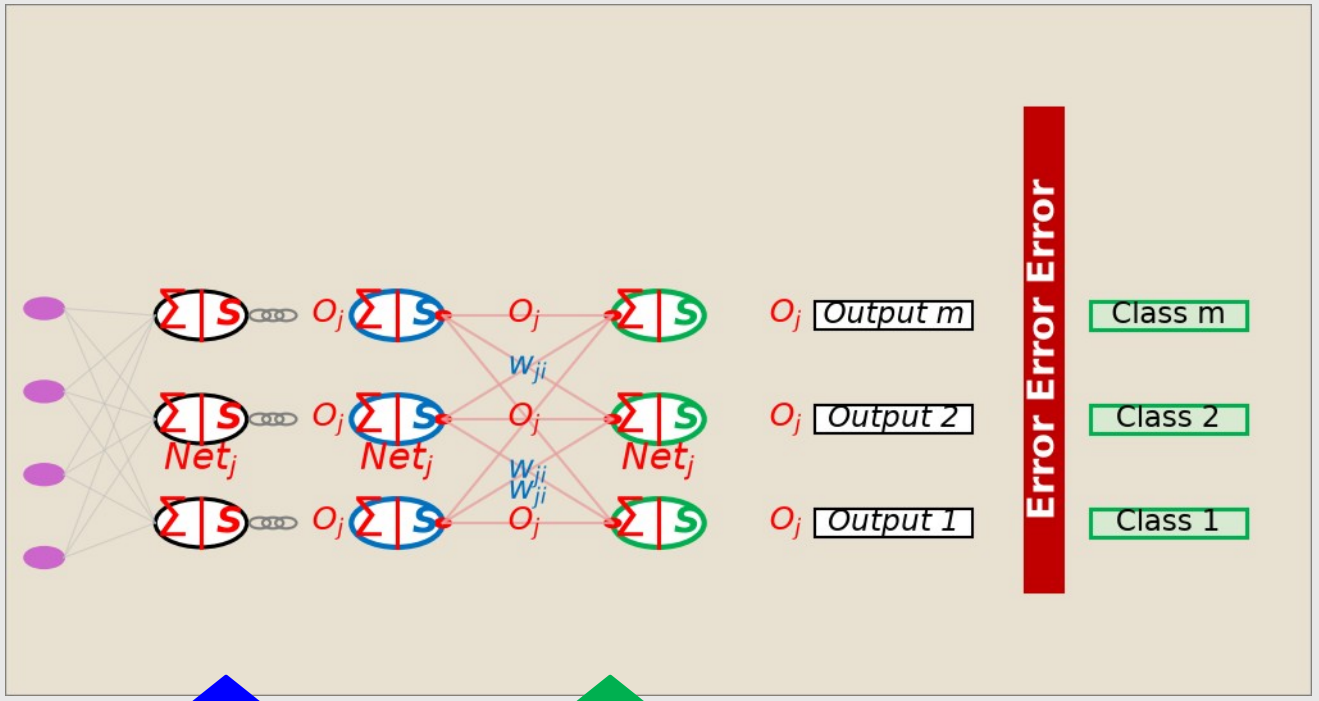
Neural Network Model

InputSample → **TargetLabel**



Error Error Error

- Class 1
- Class 2
- Class m

Net_j ⇒ Summation at Node j

O_j ⇒ Output at Node j

w_ji ⇒ Weight from Node i to Node j

Number of I/P Nodes = Input Vector Length
Number of O/P Nodes = Number of classes

Previous LayerLayer (i)

Current Layer(You are here)

For Layer "j"

$$Net_j = \sum_i w_{ji} O_i$$

$$O_j = f(Net_j) = \frac{1}{1 + e^{-Net_j}}$$

If Layer "j" is Output Layer:

$$E = \frac{1}{2} \sum_j (t_j - O_j)^2$$